

Point Cloud Sequence Encoding for Material-conditioned Graph Network Simulators

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1 The problem

- Learned simulators** (graph network simulators) are fast and differentiable, but must be told the **material**: stiffness, damping, thickness ...
- In-context learning** (MaNGO) infers the material from a few example trajectories, but needs **mesh trajectories**, which real sensors do not provide.
- Depth cameras** give point clouds, with **no correspondences** between frames.

Can a simulator adapt to a new material from a few **point cloud recordings**, without any optimization loop?

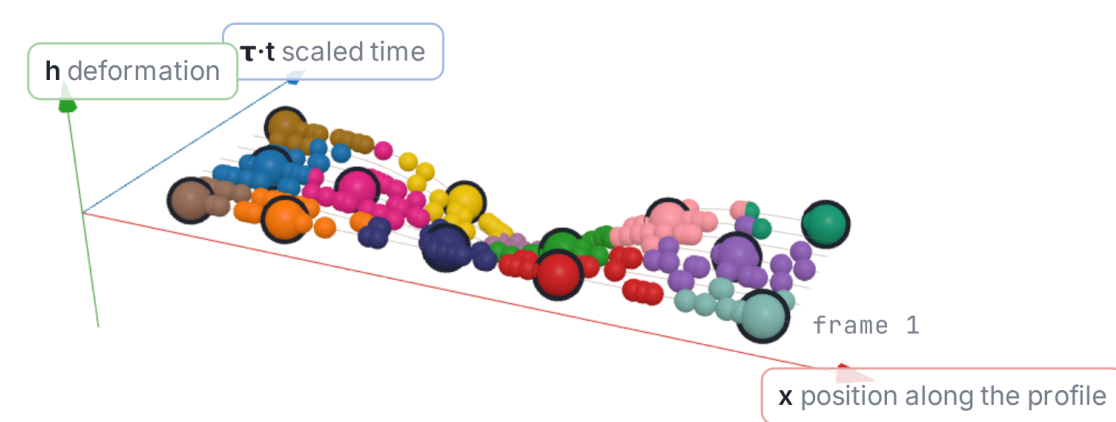
2 How PEACH works

- Observe** 1–8 point cloud sequences of the target process, e.g. with a depth camera.
- Encode** each sequence as **one point cloud in 4D space-time**: FPS + KNN patches, a PointMAE tokenizer on Fourier features and one transformer layer.
- Aggregate** the sequence codes with a learned softmax into one **latent material description** r .
- Simulate** the **whole trajectory in one forward pass** with the MaNGO graph network simulator, conditioned on r and an initial mesh.

Trained end-to-end in simulation, with point cloud augmentation (noise, region dropout, outliers) for sim-to-real.

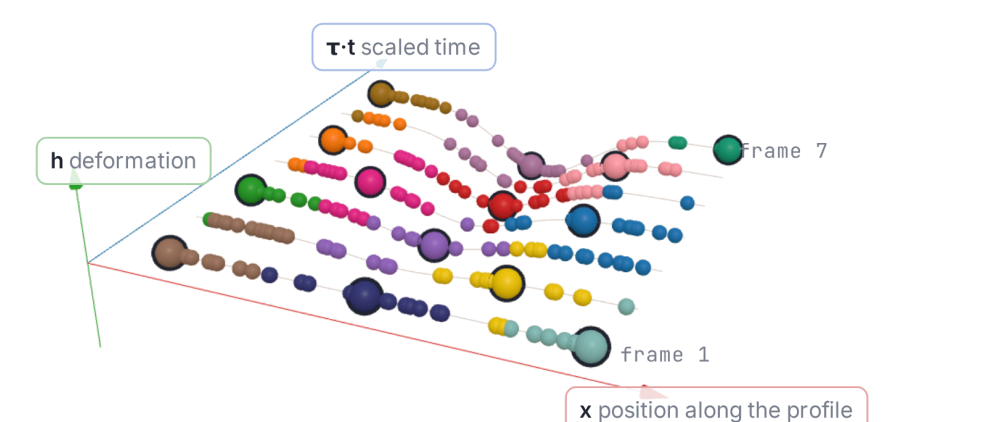
3 Space-time patches

Time is normalized and scaled by τ . Patches then group points that are close in space and time, so motion becomes visible without point correspondences.



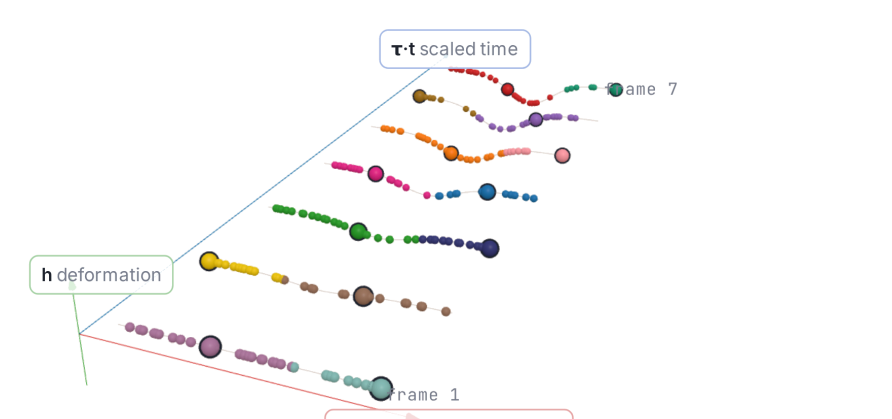
$\tau = 0.3$

time barely counts: patches mix several frames



$\tau \approx 1$

patches are local in space and time

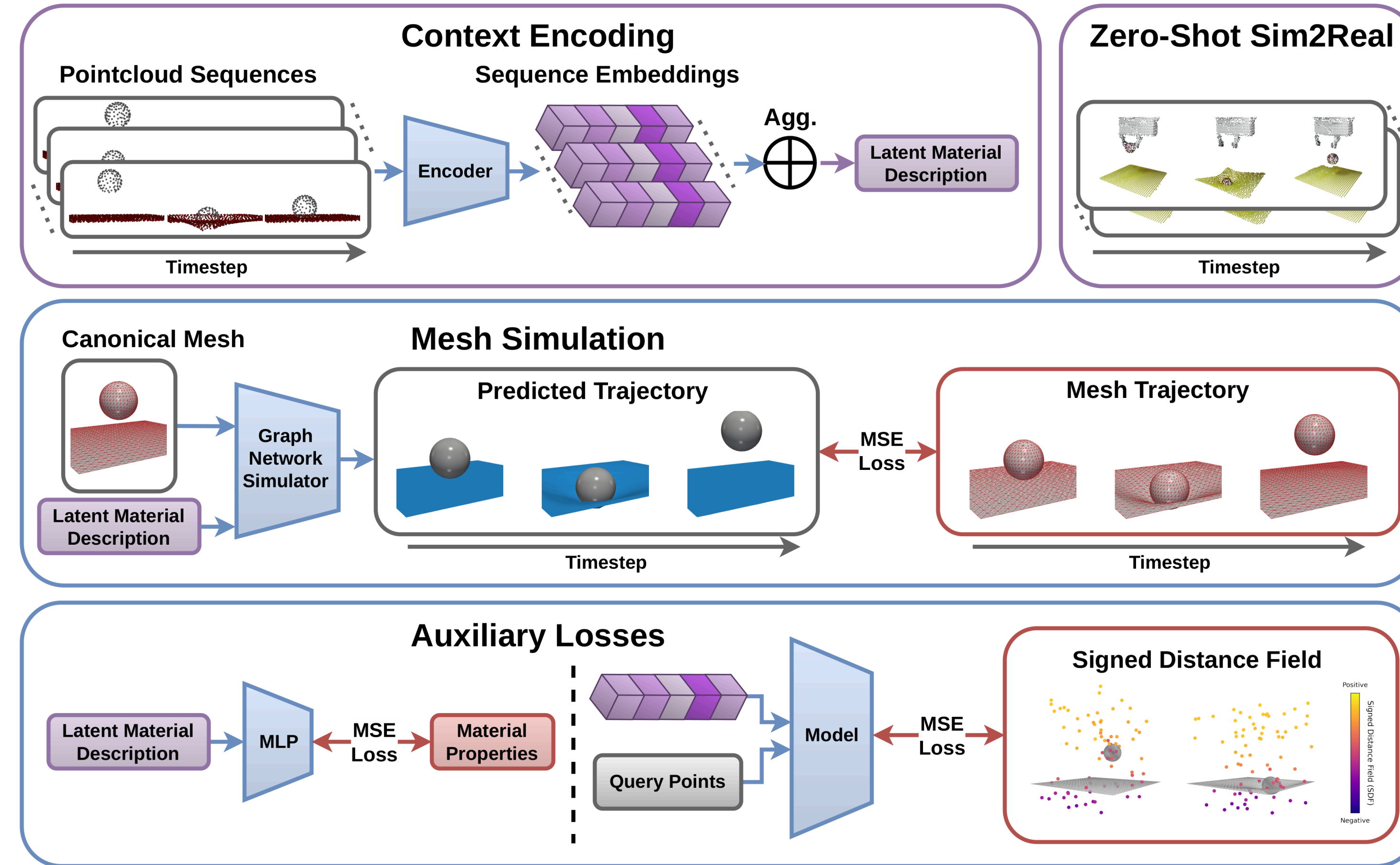


$\tau = 3$

one frame per patch: motion is invisible to a single patch

4 Auxiliary losses (training only)

- Material regression**: a small MLP predicts the physical parameters from r . It grounds the latent space and doubles as system identification.
- Signed distance field**: query points predict their distance to the mesh surface from nearby patch tokens. This keeps geometric detail in the tokens and needs geometry only.



1–8

context point cloud sequences plus one initial mesh

0

real point clouds in training zero-shot sim-to-real

45%

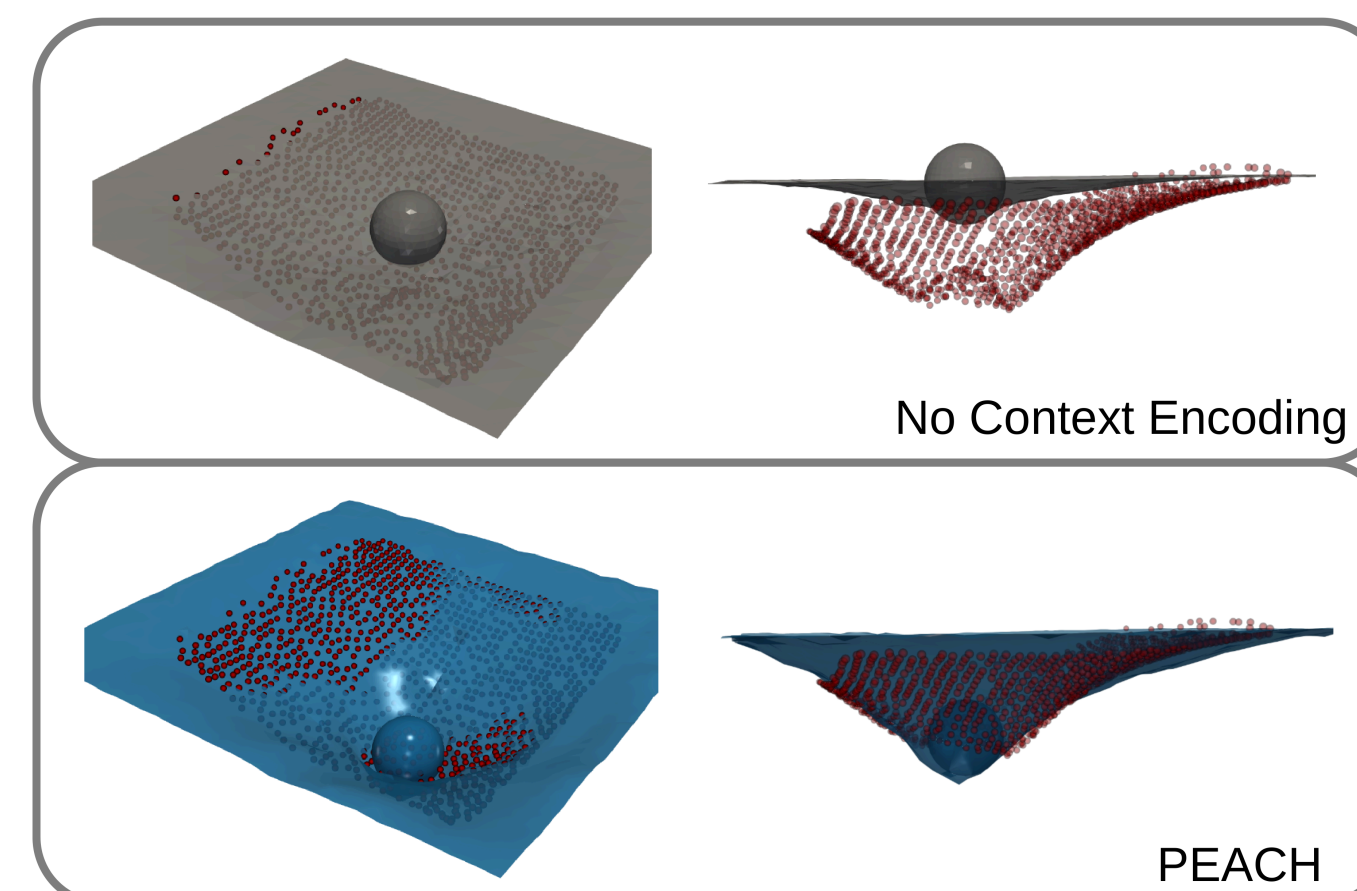
lower real-world error than the Oracle at maximal sheet stretching

~140×

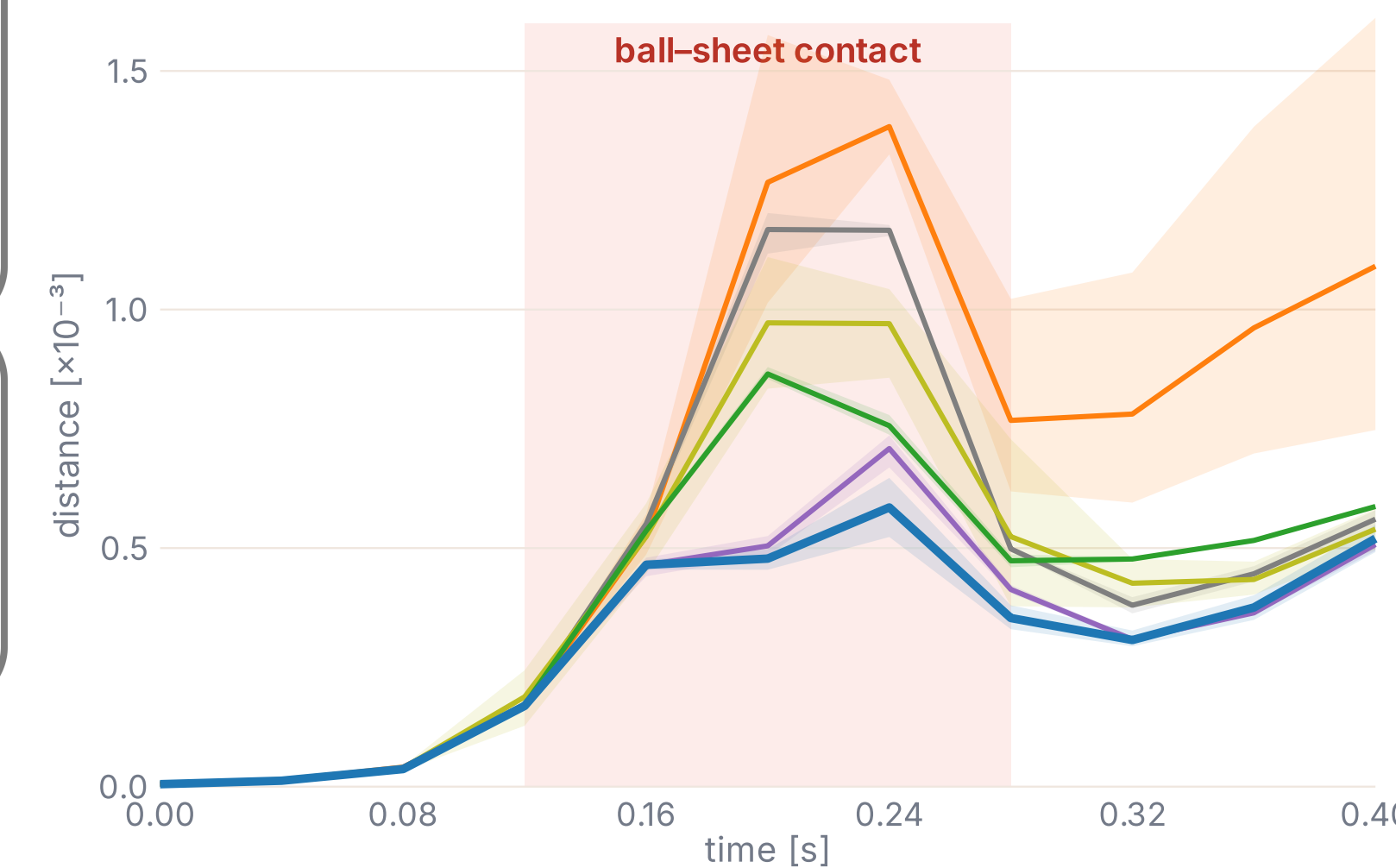
faster than test-time optimization 0.58 s vs. 82 s

5 Zero-shot sim-to-real

A robot drops balls of different size and weight onto latex sheets of different thickness. Only stereo-camera point clouds are recorded; PEACH was trained purely in simulation.

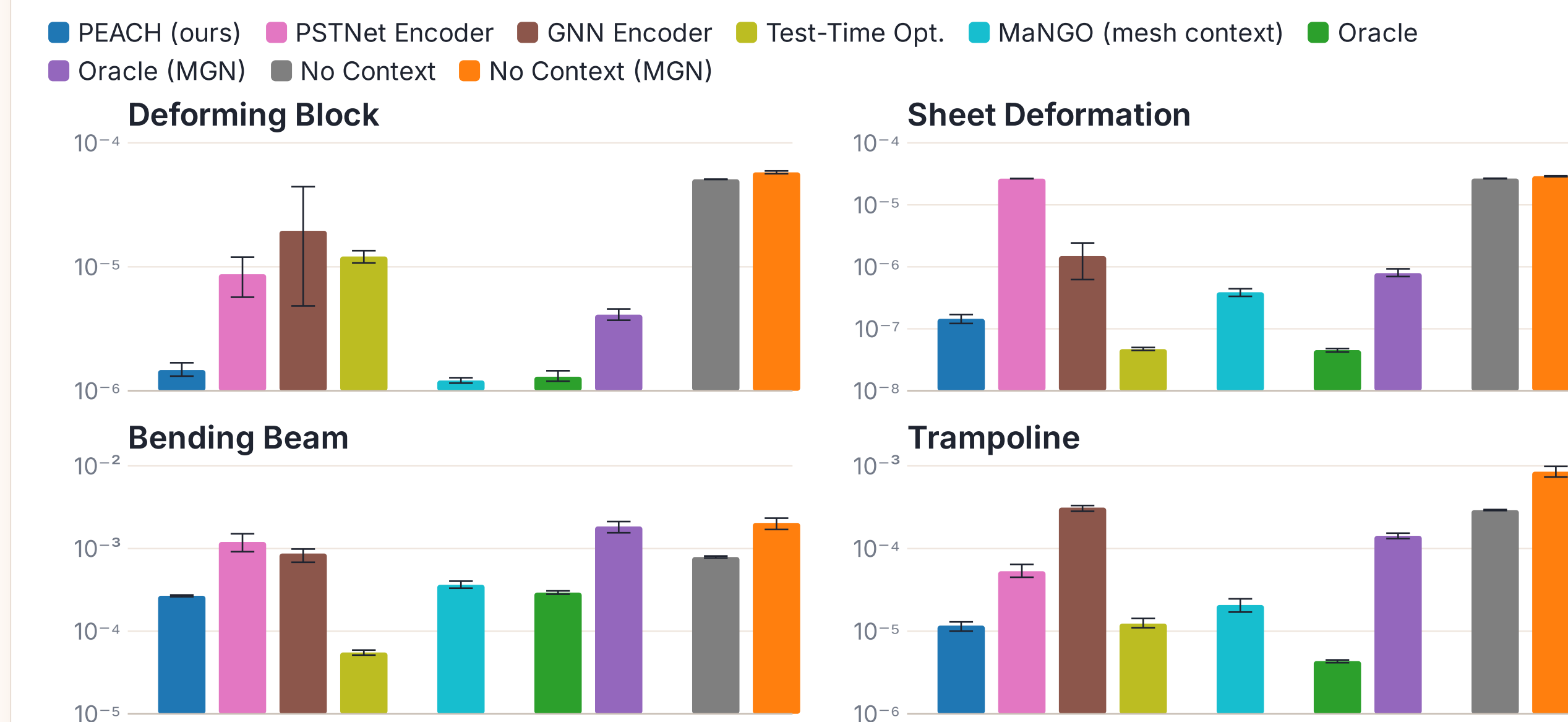
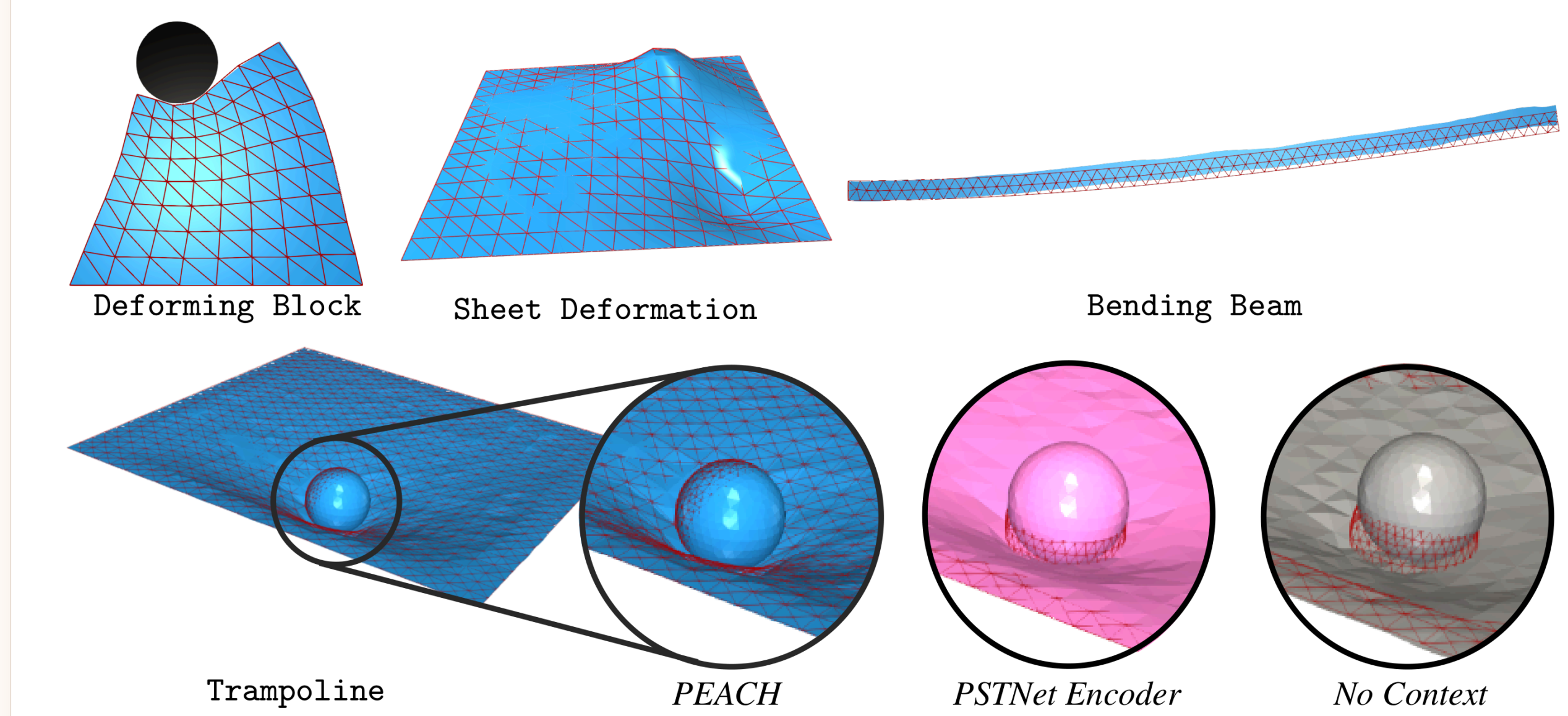


At maximal stretching ($t = 0.2$ s): the context-free simulator barely deforms, PEACH follows the observed point cloud (red).



■ PEACH (ours) ■ PEACH, no data augm. ■ "Oracle" ■ Test-Time Opt. ■ No Context ■ No Context (MGN)
Mean point-to-mesh distance over all real trials. PEACH is best during the contact phase.

6 Simulation results



Full rollout MSE with 8 context trajectories (log scale, lower is better). PEACH beats all learned point cloud encoders and is on par with mesh-based MaNGO, which uses privileged mesh context.



No optimization loop: adapting and simulating one task takes 0.58 s, vs. 82 s for test-time optimization and 240 s for one FEM rollout.

7 Structured latent space

Pairwise distances between tasks in the latent space r follow their distances in physical parameters ρ . Darker hexagons hold more task pairs.

